AARADHYA BHATIYA

Game Designer · Systems, Progression & Economy Balance

Hands-on designer who takes content from concept to live release, in data and in engine

Indore, India | +91 73899 77836 | bhatiyaaaradhya@gmail.com

Portfolio | LinkedIn | GitHub | itch.io

Summary

Hands-on game designer and developer with 3+ years shipping games, specialising in systems, progression, economy and numerical balancing. I take ideas from concept to release on my own: design, configure the content in data, implement it, playtest and iterate. Most recently I designed and built a co-op online game end-to-end, with difficulty tiers, a simulation-tuned 16-rank progression curve, an itemized reward economy, 45 scripted events and playtest-driven updates. Comfortable working inside existing codebases and toolchains, scripting in Lua/Luau, C# and SQL, and creating or adapting game art. Clear, practical documentation is part of how I work.

What I bring to this role

- **Progression, rewards and motivation:** designed a 16-rank progression curve with a new unlock almost every early session, one-time rewards that pull players toward new content instead of grinding, and itemized payouts that show players exactly what earned them what.
- **Numerical balancing and economy:** designed RTP and probability models for four shipped casino games; tuned a full progression and economy with Monte Carlo simulation; balanced difficulty through ~12 separate levers per tier, not just multipliers.
- **PvE and encounter design:** 5 enemy types with distinct behaviours and counters, an escalating 5-stage threat model, and 45 events paced by stage, with rare set pieces and quiet periods.
- **Live events and repeatable activities:** rotating side objectives, 12 location stories with randomized clue placement, and a rare 3% event with its own deduction puzzle, all driven from configuration.
- **Configuration, data and tools:** content and balance live in data tables (Lua/Luau modules, ScriptableObjects), so updates are config changes; built 36 internal editor tools across projects; comfortable inheriting and working within an existing codebase and its constraints.
- **Player feedback to updates:** run structured feedback rounds of 20–25 items each, triage and ship them, and document every change, e.g. re-pacing the whole game when players said too much happened too fast.
- **Art and technical art:** pixel art, Blender and custom HLSL shaders, so I can create, adapt and integrate assets myself.

Skills

Design	Systems design, progression and reward design, game economy, numerical balancing, PvE encounter and pacing design, difficulty tuning, live events, GDDs, design specs and change logs
Data & Scripting	Lua/Luau, C#, SQL, Python, JavaScript; data-driven configuration tables; probability and RTP models; Monte Carlo balance simulation; analytics events
Engines & Tools	Roblox Studio, Unity 3D/2D, GameMaker Studio, custom editor tools, Git/GitHub, JIRA, Asana, Figma, AI-assisted development (Claude Code, Roblox Studio MCP)
Art	Custom HLSL shaders, Shader Graph, VFX, Blender, Aseprite, pixel art
Live Ops	Live production title (17+ mini-games), IronSource ad mediation, analytics integration, no-pay-to-win monetization design, cross-device QA

Design Project

Paranormal Inc. Co-op online horror game | Lead designer and developer — 2026 – Present

Roblox, Luau · 1–4 players, persistent progression

- **Owned the game from concept to a playable, playtested build:** the core loop, 4 maps (80 rooms), 5 enemy types each with a unique 3-of-7 evidence fingerprint, 10 upgradeable tools with tier-2 versions, and 45 scripted events.
- **Progression:** a 16-rank reputation curve tuned by Monte Carlo simulation of ~12-minute sessions: new content after almost every early job, the hardest tier at ~16 hours, top rank at ~35 hours; map and tier unlocks staggered to keep goals in view.
- **Economy and rewards:** itemized payouts (evidence, photos, objectives, identification) with a co-op bonus of +10% per teammate, softened failure and death payouts, penalties for wrong answers, and one-time first-clear and story rewards; gear prices and reputation gates as sinks.
- **Difficulty:** 5 tiers built from separate levers (enemy aggression growth, grace period, attack thresholds, cooldowns, attack duration and speed, sanity drain, item effectiveness, hidden-information chance, spawn weights); rewards scale ×1 to ×5.
- **Pacing model:** threat level follows a baseline curve plus player-driven provocation spikes, with stage-based event cadence, quiet periods after big moments, and rare events rolled once per session.

- **Live-style content:** rotating side objectives (10 types), randomized story and clue placement, a rare blood-moon event with a deduction puzzle, an HQ hub with social activities, and fair monetization (revives and private servers only, nothing that affects balance).
- **Iteration and documentation:** multiple structured playtest rounds of 20–25 feedback items each, all triaged, shipped and documented; all balance kept in 15 data-definition modules so tuning never needs code.

Experience

Lead Game Developer TONIZTOZ | Remote — Mar 2025 – Present

The God They Buried (2D horror) · Unity 2D Demo video · toniztoz.com

- **Own the design** of an original 2D pixel-art horror game rooted in Asian folklore, from storyline, storyboard and GDD through active development, leading a five-person cross-disciplinary team.
- **Set priorities and direction** for the team, mentor juniors on Unity, Aseprite and asset workflows, and drive the game's look through custom HLSL lighting, fog and ghost shaders.

Independent Contractor Freelance | Remote — 2024 – Present

Unity, WebGL Casino showcase

- **Designed the numbers for four shipped real-time casino games** (Mines, Roulette, Crash, Teen Patti): RTP mathematical models, probability systems and payout balance, with retention-focused loops and live wallet integration (balance sync, currency switching).
- **Delivered 15+ AR projects** for clients, owning each from requirements to delivery.

Unity Developer Boop | Remote, contract — Jun 2026 – Sep 2026

3D companion app (iOS and Android) Showcase video

- **Took over an existing product mid-development** after the other developers left, learning its codebase and constraints and delivering it to a near-release build on my own.
- **Designed the behaviour system:** 27 behaviours chosen from live context by requirements and a priority ladder, with recency-weighted randomness so the character never feels repetitive; built designer-facing tools so new behaviours could be authored without code.

Unity Developer Intern Pythrust Technologies | Gurugram, India — Oct 2025 – Jan 2026

2MinWin, a live title by MINWIN Digital Studios 2min.win

- **Built 17+ mini-games for a live production title**, working with UI and sound design to keep proposed designs achievable in engine, with IronSource ads, analytics events and cross-device regression testing.

Other Projects

Chalo Yaar! 3D multiplayer travel game — Jan 2026 – May 2026

Unity itch.io · Trailer

- **Designed a cozy progression loop:** a drivable campervan hub, wildlife photography progression, inventory and survival stats; custom HLSL rendering and a 15 to 50+ fps optimisation pass.

SlenderAR · Lucius AR horror survival · procedural horror maze runner — 2024 – 2025

SlenderAR on itch.io · Lucius gameplay

- **Horror design in two formats:** real-world walking through an AR portal with player-tracking AI and dynamic audio; procedurally generated mazes with an enemy that adapts to the player's position.

Education

B.Tech Computer Science and Engineering (spz. Gaming and Graphics) UPES, Dehradun — 2022 – 2026

CGPA 7.78